Advanced Optical Fiber Communication Formulae

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The following represents a collection of formulas accompanying the advanced optical fiber Communication lecture. Formulas are taken from *Agrawal, fiber-optic communication and Petermann, Einführung in die optische Nachrichtentechnik*. Only for internal use. Feel free to comment on missing or wrong details.

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Plane Waves

$$\underline{\vec{E}}(t) = \Re(\vec{E}e^{\mathrm{j}(kx-\omega t)}) \tag{1}$$

where:

- k as the wavenumber $k = \frac{2\pi}{\lambda}$
- β as the propagation constant(free space) $\beta = kn$
- ω as the optical frequency $\omega = 2\pi\frac{c}{\lambda} = 2\pi f$
- phase velocity $v_{phase} = \frac{\omega}{\beta}$
- group velocity $v_{group} = \frac{d\omega}{d\beta} = \frac{c}{N}$
- group velocity dispersion $GVD = \frac{d^2\omega}{d^2\beta}$

Photon Energy

$$E_{phot} = hf \tag{2}$$

with

$$h = 6.62607015 * 10^{-34} Js \tag{3}$$

Sellmeier Equation

$$n^2 = 1 + \frac{\omega_e \cdot \omega_d}{\omega_e^2 - \omega^2} - \frac{\omega_L^2}{\omega^2} \tag{4}$$

this equation is derived from the Lorentz-Oscillator Model, and deviates from the standard equation.

Absorption & Scattering

$$\underline{n} = n - j\kappa \tag{5}$$

with κ as the complex part of the refractive index.

decaying wave:

$$\underline{E}_x = \underline{E}_{x0}e^{-j\beta z - \alpha z} \tag{6}$$

with

$$\alpha(\omega) = \frac{2\omega\kappa(\omega)}{c} \tag{7}$$

Maxwell Equations

$$\nabla \times \underline{\vec{E}} = -j\omega\mu_0\underline{\vec{H}} \tag{8}$$

$$\nabla \times \underline{\vec{H}} = j\omega\varepsilon\underline{\vec{E}} = j\omega\varepsilon_0 n^2\underline{\vec{E}} \tag{9}$$

where:
n as the refractive Index $n = \sqrt{\epsilon_r\mu_r}$

$$\Delta\underline{\vec{E}} + k_0^2 n^2 \underline{\vec{E}} = 0 \tag{10}$$

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waves at refractive Index Boundaries

plane wave, no polarization(Snellius):

$$n_1 \sin\varphi_1 = n_2 \sin\varphi_2 \tag{11}$$

boundary for TIR

$$sin(\varphi) = \frac{n_2}{n_1} \tag{12}$$

depth of an evanescent field @ TIR

$$d_{eva} = \frac{\lambda}{2\pi\sqrt{n_1^2 \sin\varphi - n_2^2}} \tag{13}$$

Polarized Waves
Power reflection coefficients:

$$R_s = \left| \frac{n_1 \cos\theta_1 - n_2 \cos\theta_2}{n_1 \cos\theta_1 + n_2 \cos\theta_2} \right|^2 \tag{14}$$

$$R_p = \left| \frac{n_1 \cos\theta_2 - n_2 \cos\theta_1}{n_1 \cos\theta_2 + n_2 \cos\theta_1} \right|^2 \tag{15}$$

$$T = 1 - R \tag{16}$$

Brewster Angle

$$\tan\varphi_1 = \frac{n_2}{n_1} \tag{17}$$

Fiber Parameters

$$\sin\gamma_{1g} = n_1 \sin\theta_{1g} = \sqrt{n_1^2 - n_2^2} = A_N \tag{18}$$

with A_n as the numerical Aperture
γ_{1g} free space angle
θ_{1g} fiber angle

Propagation Constant in fibers

$$\beta = k_0 n_1 \cos\theta_1 \tag{19}$$

Fiber Parameter

$$V = k_0 a\sqrt{n_1^2 - n_2^2} = k_0 \cdot a \cdot A_N \tag{20}$$

a as the Core Radius
Effective refractive Index

$$n_{eff} = \sqrt{n_2^2 + B(n_1^2 - n_2^2)} \tag{21}$$

Normalized Propagation constant (Fiber only)

$$B = \frac{\frac{\beta^2}{k_0^2} - n_2^2}{n_1^2 - n_2^2} \approx \frac{\frac{\beta}{k_0} - n_2}{n_1 - n_2} \tag{22}$$

Core Parameter

$$u = V\sqrt{1 - B} = a\sqrt{k_0^2 n_1^2 - \beta^2} \tag{23}$$

Cladding Parameter

$$v = V\sqrt{B} = a\sqrt{\beta^2 - k_0^2 n_2^2} \tag{24}$$

Characteristic Fiber equation

$$-\frac{u \cdot J_{l+1}(u)}{J_l(u)} + \frac{v \cdot K_{l+1}(v)}{K_l(v)} = 0 \tag{25}$$

Where K is the Modified Bessel Function of second kind, l_{th} order
and J is the Bessel function of first kind.

$$V^2 = u^2 + v^2 \tag{26}$$

Marcuse's approximation & modal confinement

for first order mode only

$$\frac{\omega}{a} \approx 0.65 + 1.619V^{-3/2} + 2.879V^{-6} \tag{27}$$

$$\Gamma = 1 - exp\left[-\frac{2a^2}{w^2}\right] \tag{28}$$

Coupling of Fibers

SMF coupling efficiency

$$\eta = \left(2 \cdot \frac{w_1 \cdot w_2}{w_1^2 + w_2^2}\right)^2 \exp\left(-2 \cdot \frac{\Delta x^2}{w_1^2 + w_2^2}\right) \tag{29}$$

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relative fiber delay

$$\tau_t = \frac{n_1 - n_2}{c}\frac{d(VB)}{dV} \tag{30}$$

absolute delay t

$$t = \tau_t \; L_{oa} \tag{31}$$

Fiber Dispersion

$$D = -D_{waveguide} - D_{Material} \tag{32}$$

$$D = -\frac{n_1 - n_2}{c\lambda} V \frac{d(VB)^2}{d^2V} - \frac{\lambda}{c}\frac{dn_2^2}{d^2\lambda} \tag{33}$$

$$D_{Material} = -\frac{2\pi c}{\lambda^2}\beta_2 \tag{34}$$

with $\beta_2 = \frac{d^2\beta}{d\omega^2}$

Approximation of $D_{Material}$

$$D_{Material} = 122\frac{1 - \lambda_{ZD}}{\lambda} \tag{35}$$

Empiric Formula, for of $1.25\mu m < \lambda < 1.66\mu m$, standard $\lambda_{ZD} = 1310nm$

Gaussian Pulse

Pulse length(only Gaussian Pulses)

$$T_{FWHM} = 2\ (ln2)^{1/2}\ T_0\ \approx 1.667\ T_0 \tag{36}$$

RMS

$$\sigma_0 = \frac{T_0}{\sqrt{2}} \approx \frac{T_{FWHM}}{2.355} \tag{37}$$

Pulse Broadening

$$\frac{T_1}{T_0} = \left[\left(1 + \frac{C\ \beta_2\ L_{oa}}{T_0^2}\right)^2 + \left(\frac{\beta_2\ L_{oa}}{T_0^2}\right)^2\right]^{1/2} \tag{38}$$

with C as the chirp factor
agglomerated Dispersion

$$\sigma_{tot} = \sqrt{\sigma_0^2 + \sigma_D^2} \tag{39}$$

with

$$\sigma_D = |D|\ \sigma_\lambda\ L_{oa} \tag{40}$$

and the wavelength range

$$\sigma_\lambda = \frac{\lambda^2}{4\pi\ c\ \sigma_0} \tag{41}$$

Maximum Symbol-rate

$$\frac{T_B}{4} \geq \sigma \tag{42}$$

(only applicable in pulse broadening calculations)

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optical Signal to noise ratio for amplified links

$$\begin{aligned} OSNR_{dB} = 58\ dB\ + \\ 10\ log_{10}\ (\frac{P_0}{1mW}) - 10\ log_{10}(F_{amp}) \\ -\ \alpha_{dB} L_{oa} - 10\ log_{10}\ (\frac{L}{L_{oa}}) \end{aligned} \tag{43}$$

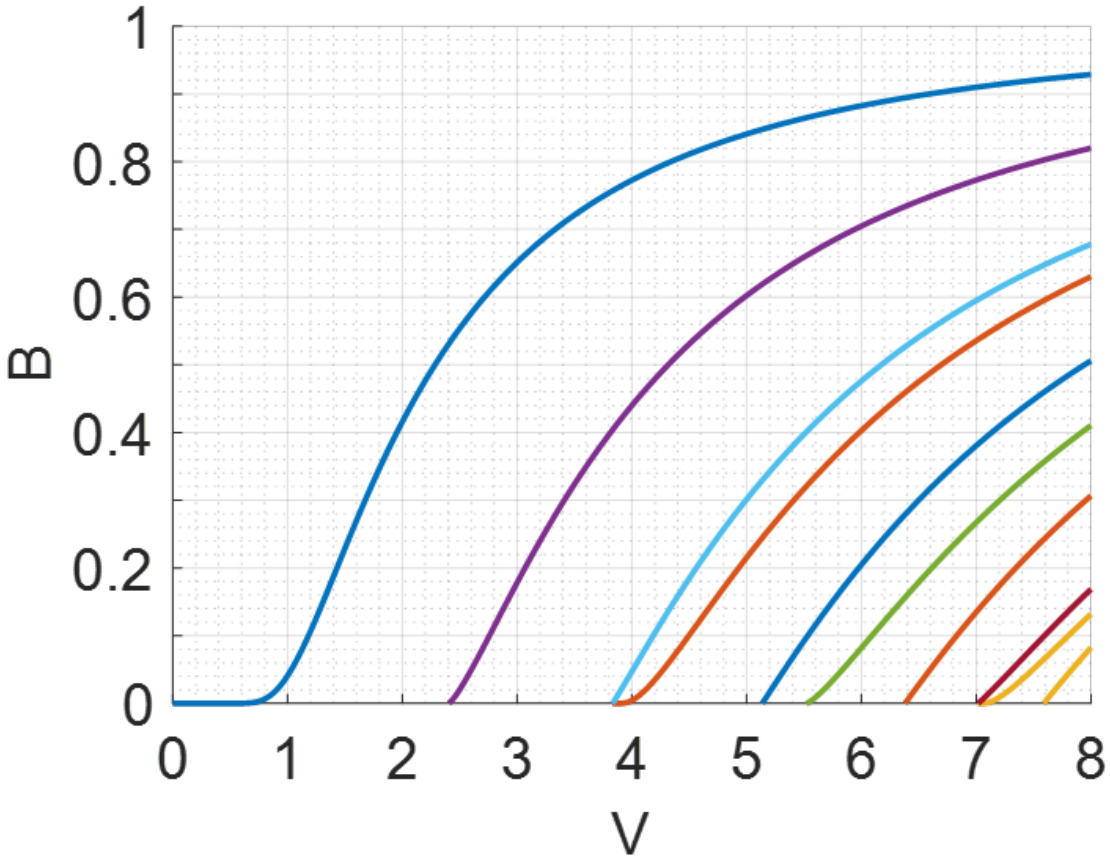
M-QAM bitrate

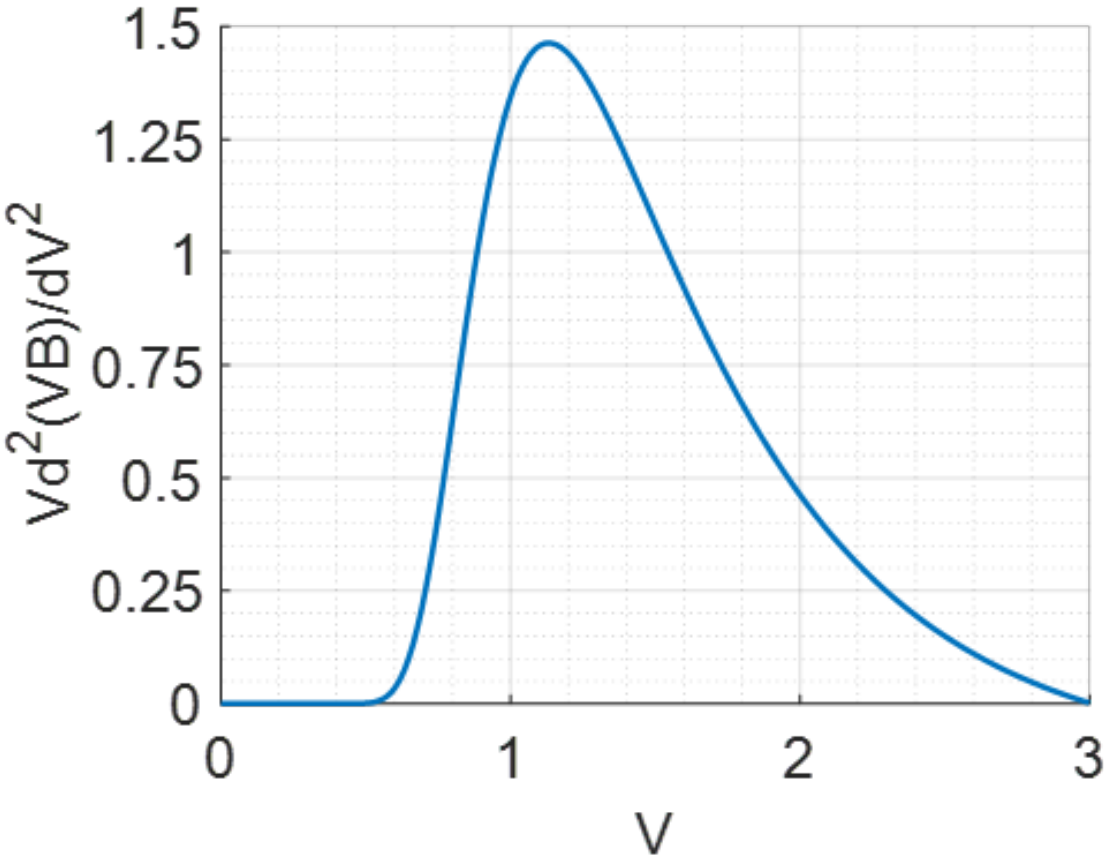
$$R_b = N_{pol}\ \frac{R_s}{1\ +\ OH}\ log_2 M \tag{44}$$

with OH as the overhead
M as the constellation order
N_{pol} polarization diversity

Shanon channel capacity limit

$$c = 2\ \Delta f\ \sum_{i=1}^{N} log_2\ (1 + \lambda_i\ SNR) \tag{45}$$


B
V
0
0.2
0.4
0.6
0.8
1
0
1
2
3
4
5
6
7
8


Vd²(VB)/dV²
V
0
0.25
0.5
0.75
1
1.25
1.5
0
1
2
3